Study B — is bacterial vaginosis more common in one arm?

The question: is BV more common in one contraceptive arm than the others?

This matters for reading yesterday's report (`02-community-composition.qmd`): BV dominated the community ordination (R^2 13%), so if BV *also* clustered in one arm, that arm would look compositionally different for a boring reason. Today we check the claim directly, with the habits from Tuesday's anatomy module: **look at the table before testing it**.

```
library(tidyverse)
library(here)

source(here("R/count_tables.R"))

dd <- read_counts("group_b")            # all six tables
dd$metadata                             # one row per sample; dd$species = the count table
```

```
# A tibble: 93 x 8
   sample_id participant_id arm        bv      nugent    ph any_sti rnaseq_batch
   <chr>     <chr>          <chr>      <chr>    <dbl> <dbl>   <dbl> <chr>
 1 B001      B001           pill       negati~      1   4.4       1 B1
 2 B002      B002           injectable negati~      0   5.3       1 B1
 3 B003      B003           injectable negati~      0   4.1       0 B2
 4 B004      B004           ring       positi~      8   5.3       0 B2
 5 B005      B005           pill       positi~      8   4.4       1 B1
 6 B006      B006           ring       negati~      0   5         1 B1
 7 B007      B007           ring       negati~      0   4.4       1 B1
 8 B008      B008           injectable positi~      8   5.3       0 B2
 9 B009      B009           injectable negati~      0   5.6       0 B1
10 B010      B010           pill       negati~      2   5.3       1 B1
# i 83 more rows
```

```r
md <- dd$metadata
```

1. Look: what does the `bv` column actually contain?

The anatomy module's first lesson: check what a column holds before you trust it. `bv` has three levels; `nugent` is the numeric score (0–10) that BV categories are conventionally derived from: 0–3 negative, 4–6 intermediate, 7–10 positive. If the two columns disagree, one of them is wrong and every downstream result inherits the error.

```r
table(bv = md$bv,
      nugent_category = cut(md$nugent, c(-Inf, 3, 6, Inf),
                            labels = c("nugent 0-3", "nugent 4-6", "nugent 7-10")))
```

```
              nugent_category
bv             nugent 0-3 nugent 4-6 nugent 7-10
  intermediate          0          3           0
  negative             51          0           0
  positive              0          0          39
```

Perfect agreement: every `negative` sits in 0–3, every `intermediate` in 4–6, every `positive` in 7–10. Two sources, one story — the column is safe to use.

2. Look: BV by arm

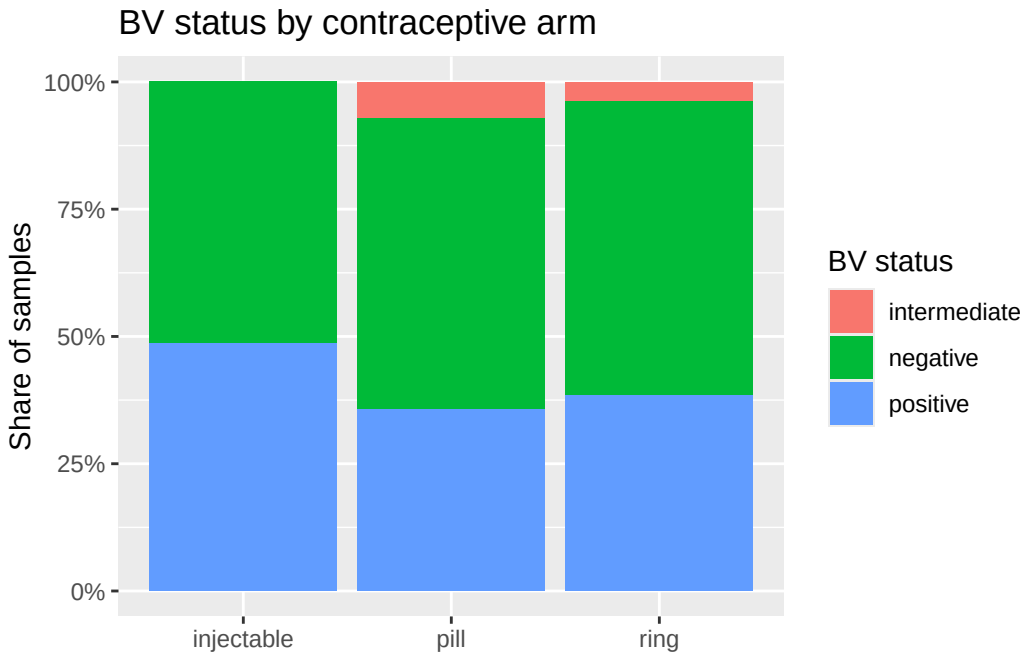
```r
# The contingency table: rows = arm, columns = BV category.
# prop.table(..., 1) turns counts into row proportions (share within each arm).
t3 <- table(md$arm, md$bv)
t3
```

```
             intermediate negative positive
  injectable            0       20       19
  pill                  2       16       10
  ring                  1       15       10
```

```r
round(prop.table(t3, 1), 3)
```

```
            intermediate negative positive
  injectable        0.000    0.513    0.487
  pill              0.071    0.571    0.357
  ring              0.038    0.577    0.385
```

```r
md |>
  count(arm, bv) |>
  ggplot(aes(arm, n, fill = bv)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(x = NULL, y = "Share of samples", fill = "BV status",
       title = "BV status by contraceptive arm")
```



- **Check me:** the counts must sum to 93 — `nrow(md)`. The injectable bar should have no visible "intermediate" segment (it has zero).

3. Test: is the pattern more than chance?

Fisher's exact test asks: "if BV status were assigned at random, how often would a table this lopsided appear?" We prefer it to the chi-square test here because two cells hold fewer than 5 samples, which makes the chi-square approximation unreliable (R warns about this if you run it).

```
fisher.test(t3)
```

```
    Fisher's Exact Test for Count Data

data:  t3
p-value = 0.4542
alternative hypothesis: two.sided
```

And the simpler two-category version — any sign of BV (positive or intermediate) vs none — in case the three-way split is just too fine for 93 samples:

```
fisher.test(table(md$arm, bv_any = md$bv != "negative"))
```

```
    Fisher's Exact Test for Count Data

data:  table(md$arm, bv_any = md$bv != "negative")
p-value = 0.8967
alternative hypothesis: two.sided
```

- **Check me:** rerun with `chisq.test()` in place of `fisher.test()` — the p-values are in the same ballpark (0.47 and 0.84), so the choice of test is not driving the answer.

4. A sanity check from the biology

If the `bv` labels mean anything, samples labelled BV-positive should carry more *Gardnerella*, the genus most associated with bacterial vaginosis.

```r
tax  <- dd$taxonomy
cnts <- dd$species
gardnerella_share <- colSums(cnts[tax$Genus == "Gardnerella", ]) / colSums(cnts)

md |>
  mutate(gardnerella_share = gardnerella_share[sample_id]) |>
  group_by(bv) |>
  summarise(median_gardnerella = scales::percent(median(gardnerella_share), 0.1))
```

```
# A tibble: 3 x 2
  bv           median_gardnerella
  <chr>        <chr>
1 intermediate 1.9%
2 negative     1.1%
3 positive     24.6%
```

Verdict

No. BV prevalence is statistically indistinguishable across the three arms. Counting `intermediate` as BV: about **49%** in the injectable arm, **43%** in the pill, **42%** in the ring; counting `positive` only: 49% / 36% / 38%. Fisher p 0.45 with three categories, p 0.90 with two — nowhere near significance, and with 26–39 participants per arm even the largest gap between arms (13 points, pill vs injectable, positive-only) is within what chance produces.

This closes the loop opened in `02-community-composition.qmd`: BV overwhelmingly shapes these vaginal communities (R^2 13%), but it is **evenly spread across arms**, so it is a rival explanation for any community pattern — not a confounder hiding inside one arm. The suggestive pill-vs-injectable difference from yesterday is therefore *not* explained by one arm simply having more BV.

Bottom line for the group: BV is common (~43–49%) everywhere in Study B; no arm is protected or penalised. Any arm effect on composition has to work through something other than BV.

Choices made

- Used **Fisher's exact test** as the main test because two cells had fewer than 5 samples; the chi-square alternative gave the same verdict.
- Ran the test **both ways**: three BV categories, and any-BV-vs-none collapsed to two; the alternative was to pick one and hope.

- Counted **`intermediate as BV`** in the two-category test (`bv != "negative"`); the alternative (positive-only, `bv == "positive"`) gives p 0.56 — the verdict does not change.
- **Kept all 93 samples**, including the nine low-depth ones flagged in `01-qc.qmd`: BV status comes from the Nugent score, not from sequencing depth, so shallow samples are still valid rows in this table.
- Validated the **`bv` column against `nugent`** before using it; the alternative was to trust the label, which would have been fine here but is a bad habit.